Short Report: Total Pressure Field Implementation in OpenFOAM

Fluid Dynamics Assignment

October 25, 2026

1 Introduction

The objective of this task is to compute the total pressure field p_{total} within the `pitzDaily` tutorial case using a runtime `codedFunctionObject`. The total pressure is defined mathematically as:

$$p_{\text{total}} = p + \frac{1}{2}|\mathbf{U}|^2 \tag{1}$$

Where p represents the static pressure field, and $\frac{1}{2}|\mathbf{U}|^2$ represents the dynamic pressure component (normalized by density ρ for incompressible flow simulation).

2 Implementation in `controlDict`

To avoid modifying the core solver source code, a dynamic C++ function object was written inside the `system/controlDict` file. The snippet reads both fields from the database, executes the algebraic expression on every cell volume, and writes out the resulting field at every designated output interval.

```
p_total
{
    type            coded;
    libs            (utilityFunctionObjects);
    name            p_total;
    writeControl    writeTime;

    codeWrite
    #{
        const volScalarField& p = mesh().lookupObject<
    volScalarField>("p");
        const volVectorField& U = mesh().lookupObject<
    volVectorField>("U");

        volScalarField p_total
```

```
        (
            IOobject
            (
                "p_total",
                mesh().time().timeName(),
                mesh(),
                IOobject::NO_READ,
                IOobject::AUTO_WRITE
            ),
            p + 0.5 * magSqr(U)
        );

        p_total.write();
    #};
}
```

Listing 1: Coded function object code inside controlDict

3 Results and Plotting

After running the case using `pimpleFoam`, the newly written field files `p_total` were verified in the time directories. The output results can be viewed visually in ParaView or extracted along a specific pathline to generate custom spatial plots.

Uncomment and include your plot below when ready:

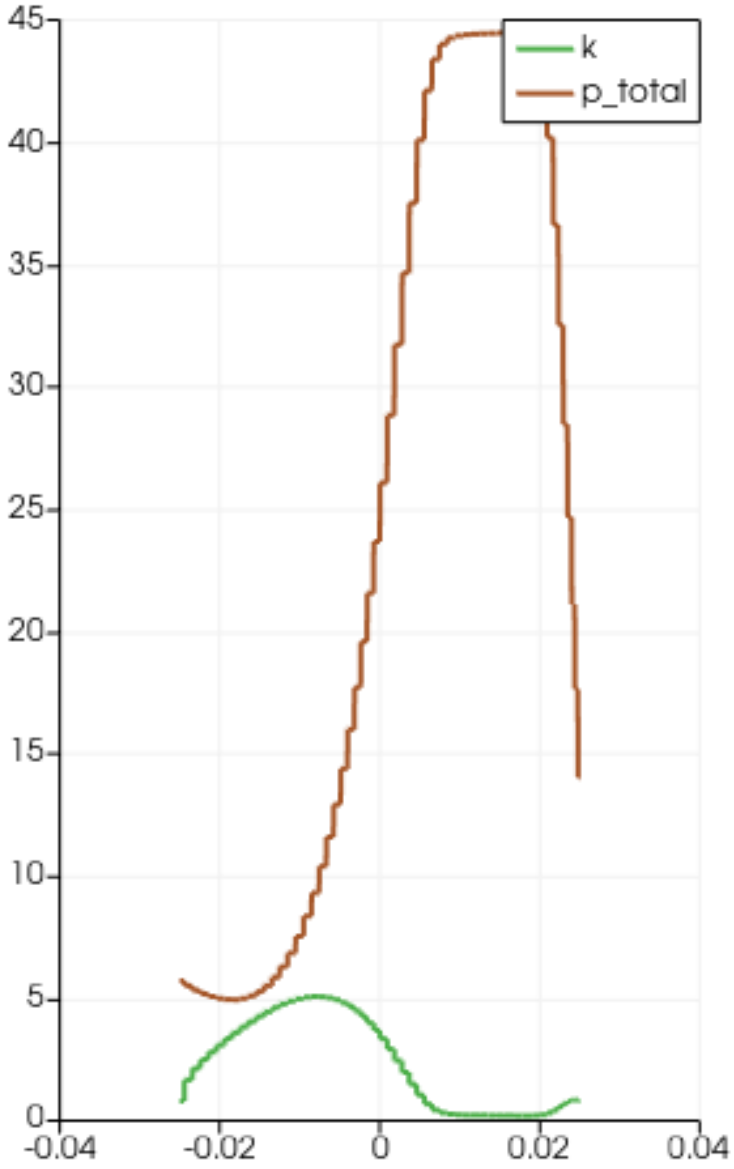


Figure 1: Total Pressure profile distribution at Time 0.3 sec across the pitzDaily channel geometry.

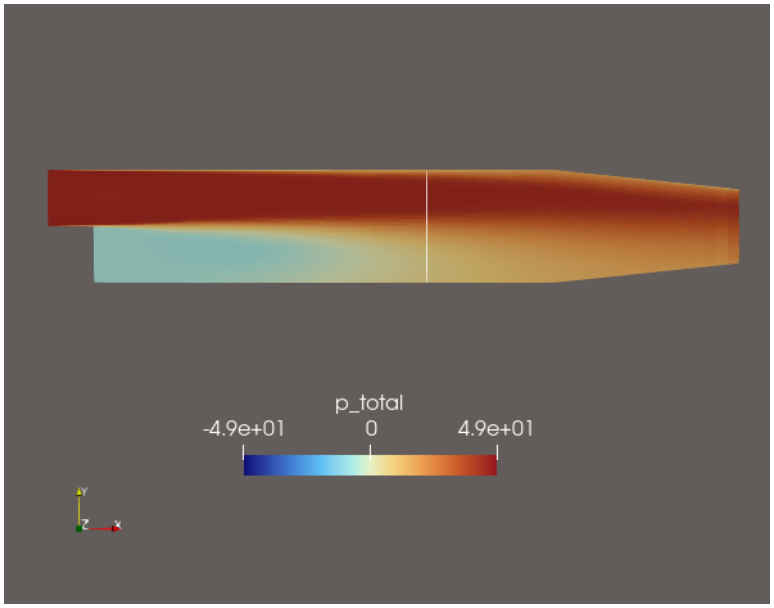


Figure 2: Total Pressure at Time 0.3 sec in the pitzDaily channel geometry.

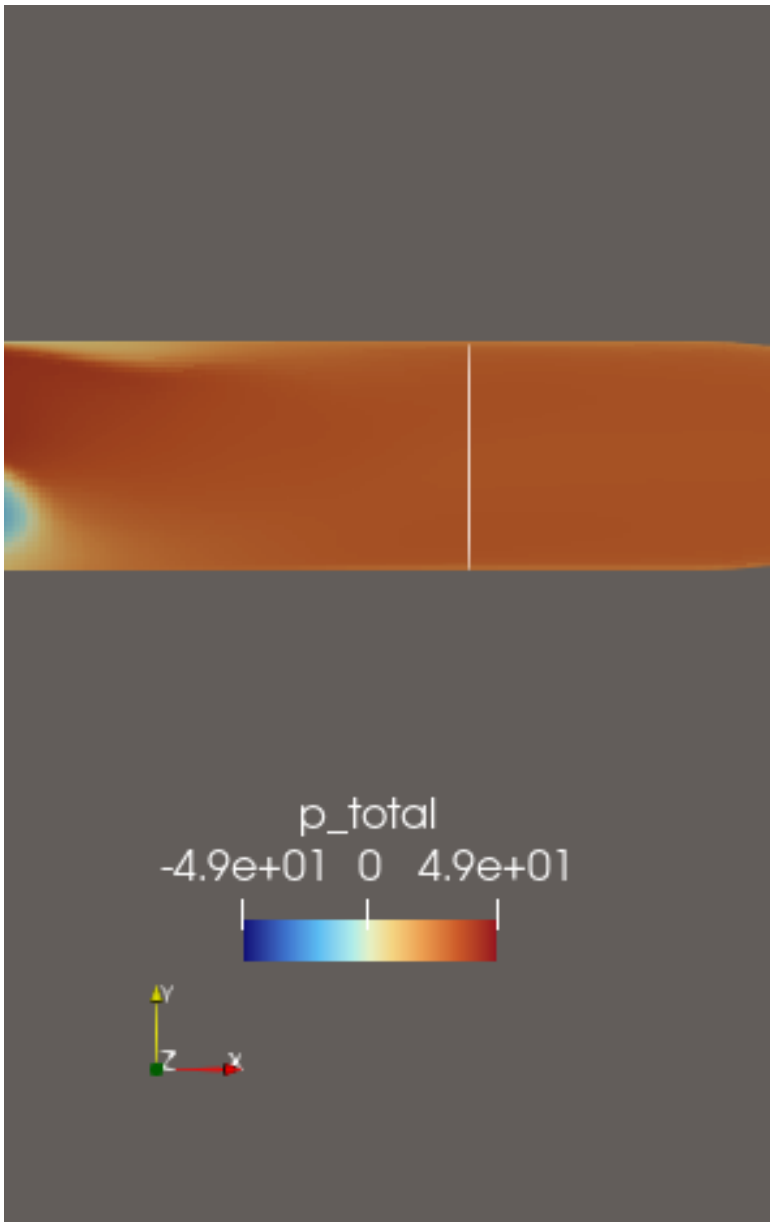


Figure 3: Total Pressure profile distribution at Time 0.02 sec across the pitzDaily channel geometry.

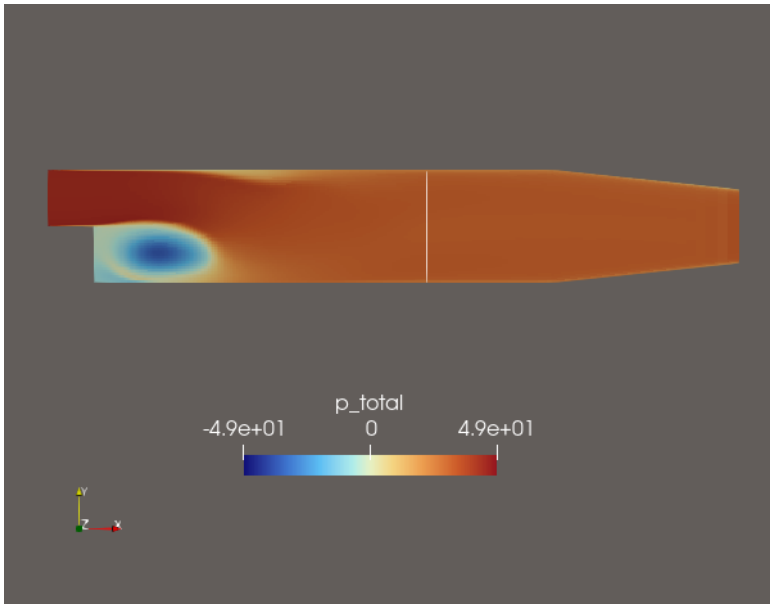


Figure 4: Total Pressure at Time 0.02 sec in the pitzDaily channel geometry.

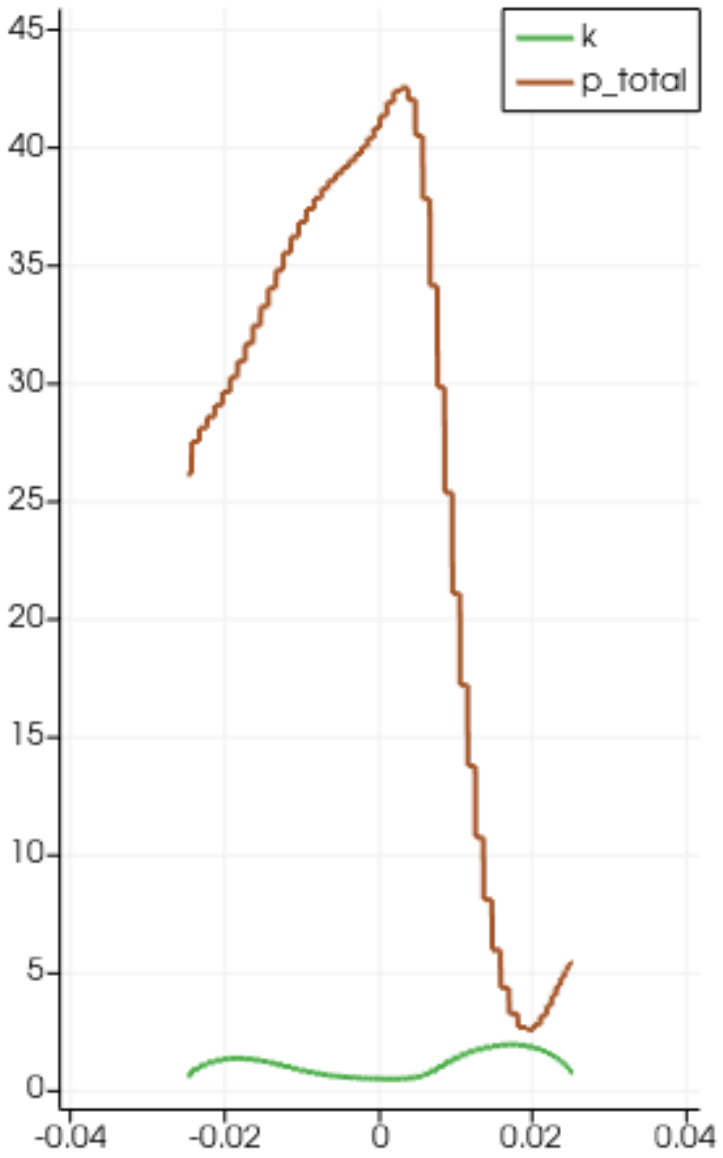


Figure 5: Total Pressure profile distribution at Time 0.05 sec across the pitzDaily channel geometry.

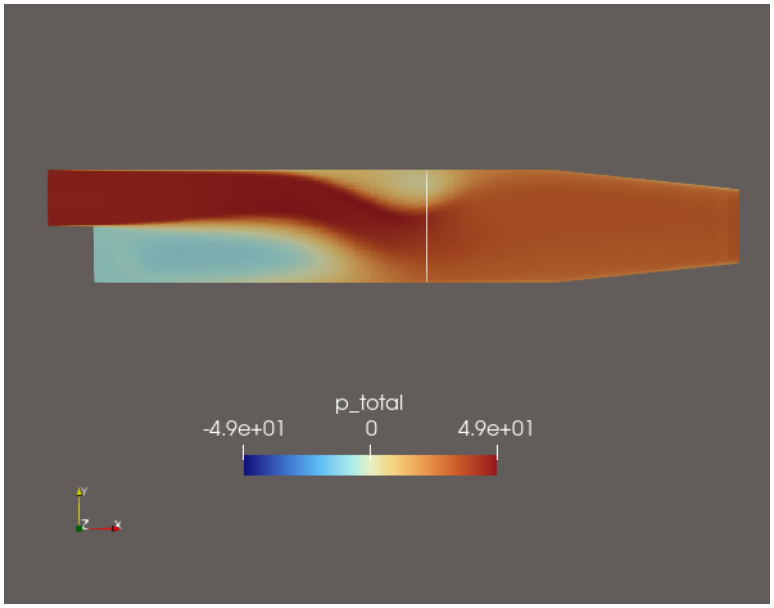


Figure 6: Total Pressure at Time 0.05 sec in the pitzDaily channel geometry.

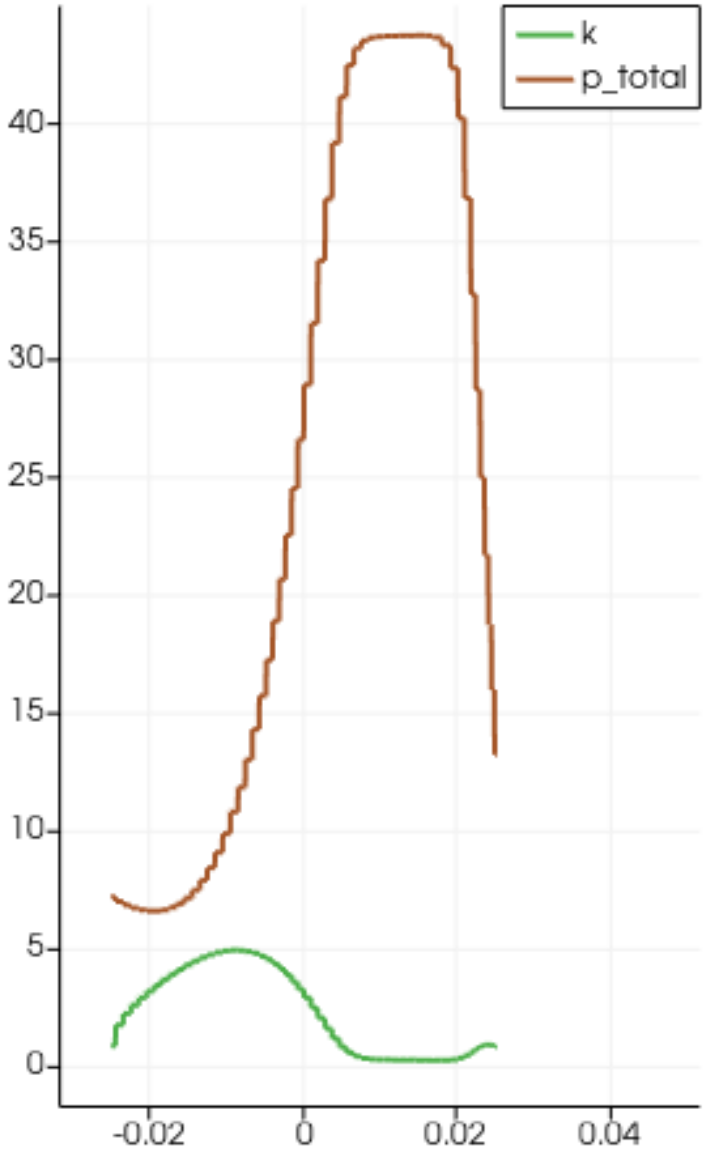


Figure 7: Total Pressure profile distribution at Time 0.1 sec across the pitzDaily channel geometry.

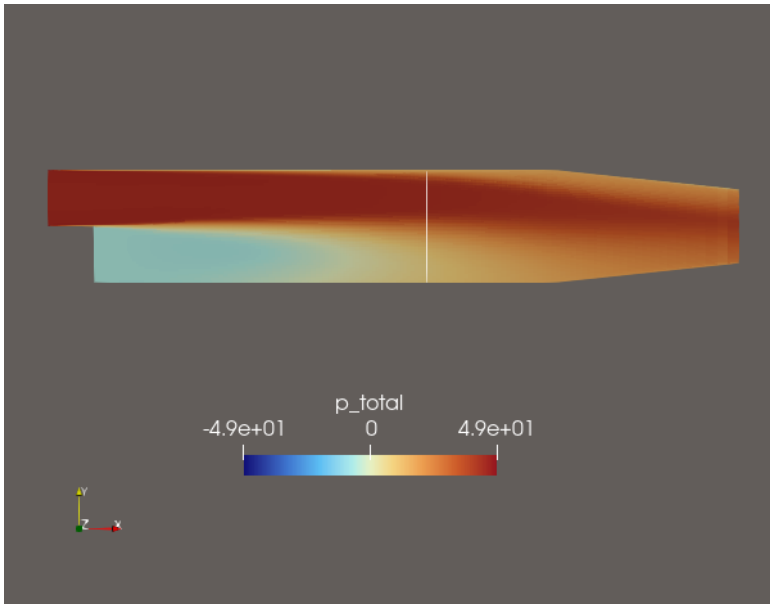


Figure 8: Total Pressure at Time 0.1 sec in the pitzDaily channel geometry.